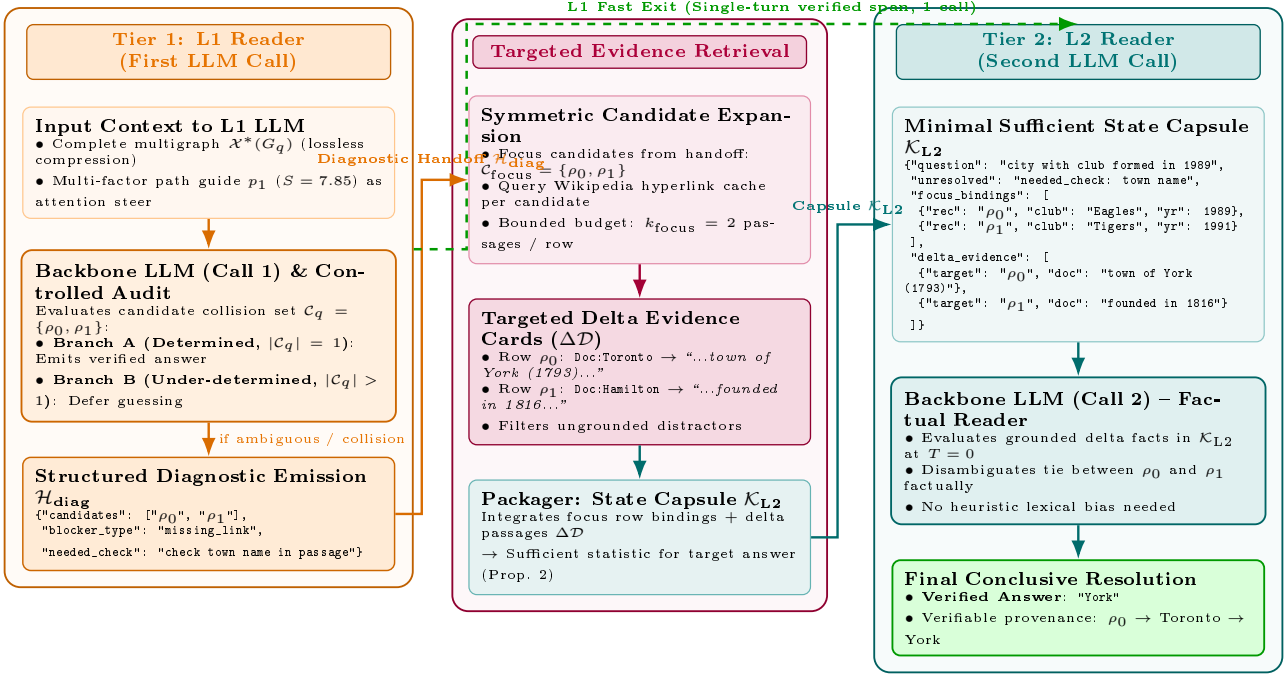

L1 Fast Exit (Single-turn verified span, 1 call)
Tier 1: L1 Reader
(First LLM Call)
Input Context to L1 LLM
• Complete multigraph $\mathcal{X}^*(G_q)$ (lossless compression)
• Multi-factor path guide $p_1$ ($S = 7.85$) as attention steer
Backbone LLM (Call 1) & Controlled Audit
Evaluates candidate collision set $\mathcal{C}_q = \{\rho_0, \rho_1\}$:
• Branch A (Determined, $|\mathcal{C}_q| = 1$): Emits verified answer
• Branch B (Under-determined, $|\mathcal{C}_q| > 1$): Defer guessing
if ambiguous / collision
Structured Diagnostic Emission $\mathcal{H}_{\text{diag}}$
{"candidates": ["$\rho_0$", "$\rho_1$"],
"blocker_type": "missing_link",
"needed_check": "check town name in passage"}
Diagnostic Handoff, $\mathcal{A}_{\text{diag}}$
Targeted Evidence Retrieval
Symmetric Candidate Expansion
• Focus candidates from handoff: $\mathcal{C}_{\text{focus}} = \{\rho_0, \rho_1\}$
• Query Wikipedia hyperlink cache per candidate
• Bounded budget: $k_{\text{focus}} = 2$ passages / row
Targeted Delta Evidence Cards ($\Delta\mathcal{D}$)
• Row $\rho_0$: Doc:Toronto → "...town of York (1793)..."
• Row $\rho_1$: Doc:Hamilton → "...founded in 1816..."
• Filters ungrounded distractors
Packager: State Capsule $\mathcal{K}_{\text{L2}}$
Integrates focus row bindings + delta passages $\Delta\mathcal{D}$
→ Sufficient statistic for target answer (Prop. 2)
Capsule $\mathcal{K}_{\text{L2}}$
Tier 2: L2 Reader
(Second LLM Call)
Minimal Sufficient State Capsule $\mathcal{K}_{\text{L2}}$
{"question": "city with club formed in 1989",
"unresolved": "needed_check: town name",
"focus_bindings": [
{"rec": "$\rho_0$", "club": "Eagles", "yr": 1989},
{"rec": "$\rho_1$", "club": "Tigers", "yr": 1991}
],
"delta_evidence": [
{"target": "$\rho_0$", "doc": "town of York (1793)"},
{"target": "$\rho_1$", "doc": "founded in 1816"}
]}
Backbone LLM (Call 2) – Factual Reader
• Evaluates grounded delta facts in $\mathcal{K}_{\text{L2}}$ at $T = 0$
• Disambiguates tie between $\rho_0$ and $\rho_1$ factually
• No heuristic lexical bias needed
Final Conclusive Resolution
• Verified Answer: "York"
• Verifiable provenance: $\rho_0$ → Toronto → York